

AI-based turbulence prediction tools are also being adopted globally. Organisations such as The Weather Company, a subsidiary of IBM, are developing AI-enhanced aviation weather platforms that deliver real-time turbulence alerts, enabling pilots to proactively adjust routes or altitudes. These capabilities enhance passenger comfort and reduce the risk of injuries caused by sudden in-flight turbulence.

In addition, AI-powered fog forecasting systems support ground operations by enabling more effective scheduling in low-visibility conditions. Several airports across Asia are trialling AI models capable of predicting fog formation up to six hours in advance—well beyond the capabilities of conventional meteorological methods.

Artificial intelligence is reshaping the aviation sector by delivering advanced



CityAirbus (Courtesy: euronews.com)

solutions that enhance safety, optimise operational efficiency, and improve the passenger experience. From air traffic management and autonomous aircraft to personalised services and weather forecasting, AI is embedded across multiple layers of aviation operations.

As these technologies continue to mature, their role will expand further, enabling more informed decision-making, reducing environmental impact, and strengthening the long-term efficiency and resilience of the aviation industry. **END** 🐧

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Inflection

So why is AI suddenly at an inflection point? We have had forms of AI since the early 1900s. The term itself appeared in 1842 and was formally coined in 1956. We have seen expert systems, rule-based systems, games such as Pac-Man and chess, and early healthcare applications. But the real acceleration began around 2011 with recommendation systems and the rise of machine learning and deep learning.

Today's inflection point is driven by four clear trends:

Data explosion: We collect data, knowingly and unknowingly, from mobile phones, vehicles, applications, and sensors. We have sufficient data to experiment, and algorithms can now generate synthetic data as well.

Cloud computing: Earlier, computing meant connecting to a mainframe somewhere else, often with delays and limitations. Today, a few clicks provide 4GB, 8GB, or 32GB of

GPU power instantly. Cloud platforms have made high-end computing accessible to every developer.

Improved algorithms: New AI models appear constantly. But the real reason for this speed is that everything is open source. This is no longer enterprise-controlled. Developers everywhere contribute to global progress.

Open contribution: The open source ecosystem is what accelerates this industry. Every improvement becomes available to everyone. This is why the field feels like it is moving at lightning speed.

Closing

AI is everywhere. AI is everything. But unless we define our own box, we

will be overwhelmed. No organisation progresses linearly, and no developer can afford to work without understanding the business and refinement cycle. AI demands a different way of thinking—structured, strategic and continuously evolving. **END** 🐧

This article is based on insights from the AI DevCon session 'Beyond Automation - Strategic AI integration with GenAI and LLM' by Pradeeba P, Delivery Manager at Thoughtworks. She comes with more than 16 years of experience delivering AI and digital solutions that drive measurable business value for global customers, and has completed the AI Strategies for Leaders program at Northwestern Kellogg School of Management. The discussion was transcribed and curated by Apurba Sen, Senior Journalist, EFFY.

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